

The CDK image of improper partition matrices and independent pair orientations

EULER

1 The original question

Chern and Fu [2, Question 5.5] ask which inversion sequences arise from improper partition matrices under the Claesson–Dukes–Kubitzke (CDK) bijection. We first give the complete value-interval answer, including the construction of the bijection. We then describe all partition matrices by allowing the unequal pairs in this answer to choose their directions independently.

A *partition matrix* on $[n] = \{1, \dots, n\}$ is an upper-triangular square matrix of sets whose nonempty entries partition $[n]$, with no empty row or column, such that

$$i < j \implies \text{col}_P(i) \leq \text{col}_P(j).$$

Here $\text{row}_P(i)$ and $\text{col}_P(i)$ are the row and column containing the label i . If consecutive labels $i, i+1$ lie in the same column, their pair is an *ascent* when $\text{row}_P(i) < \text{row}_P(i+1)$ and a *descent* when $\text{row}_P(i) > \text{row}_P(i+1)$. Such an ascent or descent is *proper* when i has the same parity as the least label in that column [2, Definition 2.1]. A matrix is *improper* if it has no proper ascent and no proper descent. Write PM_n and IPPM_n for these two classes.

An *inversion sequence* is a sequence $e = (e_1, \dots, e_n)$ with $0 \leq e_i < i$ for every i . For a partition matrix with D columns, of sizes $m_1, \dots, m_D$, the CDK map [1] is

$$\Pi(P)_i = \sum_{c < \text{row}_P(i)} m_c. \quad (1)$$

For any inversion sequence, list its distinct values and append n :

$$0 = a_1 < a_2 < \dots < a_D < a_{D+1} = n. \quad (2)$$

In each interval of positions

$$\{a_c + 1, \dots, a_{c+1}\},$$

pair the first two positions, then the next two, and so on. If the interval has odd length, leave its final position unpaired. We call these the *matched pairs* of e .

Theorem 1 (The CDK image; cf. [3]). *For every $n \geq 1$, the map (1) is a bijection from PM_n to the inversion sequences of length n . Its restriction to IPPM_n has image exactly the sequences whose entries agree at every matched pair. Explicitly, the criterion is*

$$e_{a_c+2j-1} = e_{a_c+2j} \quad \left(1 \leq c \leq D, \quad 1 \leq j \leq \left\lfloor \frac{a_{c+1} - a_c}{2} \right\rfloor \right). \quad (3)$$

Proof. Start with a partition matrix. The label-order condition and nonempty columns imply that the labels in column c form a consecutive interval. Put $a_1 = 0$ and $a_{c+1} = m_1 + \dots + m_c$. Column c therefore contains exactly $a_c + 1, \dots, a_{c+1}$. If label i lies in row r and column c , upper triangularity gives $r \leq c$, and thus

$$0 \leq \Pi(P)_i = a_r \leq a_c < i.$$

So (1) produces an inversion sequence. All the a_r are distinct because columns are nonempty, and every a_r occurs as a sequence value because row r is nonempty. Consequently its distinct values recover precisely the column endpoints in (2).

Conversely, given an inversion sequence with distinct values as in (2), put label i in row r when $e_i = a_r$ and in column c when $a_c < i \leq a_{c+1}$. Each label is assigned once. Every row is nonempty because each distinct value occurs, and every column is nonempty because successive endpoints differ. The column order of labels is automatic. If $r > c$, however, then

$$e_i = a_r \geq a_{c+1} \geq i,$$

contradicting the inversion-sequence condition. The constructed matrix is therefore upper triangular. Its column sizes are $a_{c+1} - a_c$, so the sum in (1) for row r is a_r . This recovers e . The preceding paragraph also shows that this construction recovers the starting matrix. The map is a bijection.

In column c , the least label is $a_c + 1$. Hence the positions at which a proper ascent or descent can occur are exactly the matched pairs

$$(a_c + 1, a_c + 2), (a_c + 3, a_c + 4), \dots$$

The assigned sequence values increase strictly with row number. Thus a matched pair is a proper ascent or descent precisely when its two sequence values differ. It follows in both directions that the matrix is improper exactly when (3) holds. $\square$

This answers the original question by a test on the sequence alone: its set of values determines the intervals, and equality is tested inside those intervals. There is no condition across an interval boundary or at an unpaired final position. In particular, the test is vacuous for $n = 1$. This image is not the different avoidance class equinumerous with IPPM_n in [2, Theorem 1.4]; for example, $(0, 0, 0)$ satisfies (3) but has equal first and third entries. Question 5.5 allows, but does not require, a description by pattern avoidance.

2 Independent orientations

The equalities in Theorem 1 describe the matrices in which none of the matched pairs has a direction. The natural next question is what happens when unequal pairs are allowed. The answer is especially simple with the column sizes fixed, because this fixes both the matching and the set of sequence values.

Fix positive integers $m_1, \dots, m_D$ with sum n , and write $\text{PM}_{\mathbf{m}}$ for the corresponding partition matrices. Read the labels of column c in increasing order, recording their row numbers as a word $w^{(c)}$ of length m_c over $[c]$. These words describe a partition matrix exactly when every letter of $[D]$ appears in at least one word. Pair positions 1, 2, then 3, 4, and so on within each word.

Swapping the two letters of any one of these pairs defines an involution on $\text{PM}_{\mathbf{m}}$, with equal-letter pairs left fixed. An *orbit* consists of all matrices obtainable by these swaps. For its description, retain only the unordered pair of row letters at each matched pair, together with the last row letter of every odd-length column. The order of the pairs within each column is retained; repetitions within an unordered pair are allowed.

Theorem 2 (Complete description of the orbits). *Two matrices in $\text{PM}_{\mathbf{m}}$ belong to the same orbit if and only if all these unordered pairs and unpaired letters agree. Every such record in which all row letters occur determines an orbit. If a record has d unequal pairs, its orbit contains exactly 2^d matrices, one for each independent choice of their directions.*

Every orbit has a unique member with no proper descents and a unique member with no proper ascents. For $0 \leq r \leq d$, exactly $\binom{d}{r}$ members have r proper ascents and $d - r$ proper

descents. The singleton orbits are precisely the improper matrices. The swaps preserve every cell cardinality as well as the column sizes. Under the CDK bijection they are exactly the swaps of sequence entries within matched pairs.

Proof. Both letters in a pair in column c lie in $[c]$, so swapping them preserves upper triangularity. It preserves the multiplicity of each row letter in each column, and hence preserves the cardinality of each cell and the nonemptiness of all rows and columns. Thus all the swaps are well-defined. Their positions are disjoint, so they commute.

The unordered pairs and unpaired letters are unchanged by every swap. Conversely, given these records, the only choices in reconstructing the column words are the orders of the unequal pairs. Each choice uses exactly the same row letters with exactly the same multiplicities, so every choice remains a partition matrix. All 2^d choices are distinct, and any two are connected by swapping exactly those pairs whose directions differ. This proves both the completeness of the records and the claimed orbit size.

By the source parity rule, the proper ascents and descents are precisely the increasing and decreasing unequal matched pairs. Orienting every unequal pair increasingly gives the unique member with no proper descents; orienting every pair decreasingly gives the other stated representative. Choosing which r pairs increase gives $\binom{d}{r}$ members. The case $d = 0$ is exactly improperness. Finally, the column sizes and their prefix sums do not change under a swap, so (1) swaps just the two corresponding sequence entries. $\square$

The nonempty-row condition is the only constraint connecting different column words. It restricts which unordered records exist, but it never restricts their orientations. On the sequence side this distinction is equally concrete: fixing $\mathbf{m}$ fixes the distinct value set, and swaps within its intervals preserve that set. This is why fixing column sizes is the natural boundary for the construction. The involution of Chern and Fu [2, Definition 2.2], which swaps the first proper ascent or descent, is recovered by selecting the first unequal pair in each nonsingleton orbit.

Write $p(P)$ and $q(P)$ for the numbers of proper ascents and descents. For an inversion sequence, let $T(e)$ be the set of first positions of its matched pairs. If $e = \Pi(P)$, the proof of Theorem 1 gives

$$p(P) = \#\{i \in T(e) : e_i < e_{i+1}\}, \quad q(P) = \#\{i \in T(e) : e_i > e_{i+1}\}. \quad (4)$$

Thus the image of matrices with no proper ascent is given by $e_i \geq e_{i+1}$ for $i \in T(e)$, and the image with no proper descent is given by $e_i \leq e_{i+1}$ there. More generally, a bound on $p(P) + q(P)$ is exactly the same bound on the number of unequal matched pairs. Both one-sided classes strictly contain the improper class for every $n \geq 3$: start from $(0, 1, 0)$ or $(0, 0, 1)$ and, if necessary, append $3, 4, \dots, n - 1$. The added intervals are singletons, so the original proper descent or ascent remains the only one.

Corollary 3. *Let $h_{\mathbf{m},d}$ count the orbits in $\text{PM}_{\mathbf{m}}$ having d unequal pairs, and put $H_{\mathbf{m}}(s) = \sum_d h_{\mathbf{m},d} s^d$. Then*

$$\sum_{P \in \text{PM}_{\mathbf{m}}} x^{p(P)} y^{q(P)} = H_{\mathbf{m}}(x + y), \quad (5)$$

$$\#\{P \in \text{PM}_{\mathbf{m}} : p(P) = r, q(P) = t\} = \binom{r+t}{r} h_{\mathbf{m},r+t}.$$

Each one-sided class has size $H_{\mathbf{m}}(1)$, the improper class has size $H_{\mathbf{m}}(0)$, and $|\text{PM}_{\mathbf{m}}| = H_{\mathbf{m}}(2)$. Conditional on $p(P) + q(P) = d$, a uniform matrix has $p(P) \sim \text{Binomial}(d, 1/2)$ whenever the conditioning class is nonempty. All these assertions remain true after summing over column sizes or dimensions.

Proof. An orbit with d unequal pairs contributes $(x+y)^d$. Summing this identity and extracting coefficients proves the claims. $\square$

3 Counting the orbits

The orbit description also yields an exact formula. The calculation has one global step: ensuring that no row letter is missing. This step is necessary even in a small example. For column sizes $(1, 2)$, the first column is the one-letter word 1. The possible unordered pairs in the second column are $\{1, 1\}$, $\{1, 2\}$ and $\{2, 2\}$, but the first would leave row 2 empty. Hence $H_{(1,2)}(s) = 1 + s$, rather than $2 + s$.

For $k \geq 0$, define

$$L_k(s) = k + \binom{k}{2}s.$$

This counts a single unordered pair from a k -letter alphabet: there are k equal pairs and $\binom{k}{2}$ unequal ones. For a set $S \subseteq [D]$ of forbidden row letters, put

$$k_c(S) = |[c] \setminus S|.$$

Proposition 4. *For every positive column-size vector $\mathbf{m}$,*

$$H_{\mathbf{m}}(s) = \sum_{S \subseteq [D]} (-1)^{|S|} \prod_{c=1}^D k_c(S)^{m_c \bmod 2} L_{k_c(S)}(s)^{\lfloor m_c/2 \rfloor}, \quad (6)$$

with the convention $0^0 = 1$.

Proof. If the row letters in S are forbidden, column c has an alphabet of size $k_c(S)$. Each of its $\lfloor m_c/2 \rfloor$ unordered pairs contributes $L_{k_c(S)}(s)$, and its final unpaired position, when present, contributes $k_c(S)$. At this stage the choices in all columns are independent. Inclusion–exclusion over missing row letters keeps exactly the records in which every row occurs. Theorem 2 identifies these records with the orbits. $\square$

For fixed dimension, write

$$H_{n,D}(s) = \sum_{\substack{m_1 + \dots + m_D = n \\ m_c \geq 1}} H_{\mathbf{m}}(s), \quad H_n(s) = \sum_{D=1}^n H_{n,D}(s).$$

Summing (6) over the positive column lengths gives the rational generating function

$$\sum_{n \geq D} H_{n,D}(s) z^n = \sum_{S \subseteq [D]} (-1)^{|S|} \prod_{c=1}^D \frac{k_c(S)z + L_{k_c(S)}(s)z^2}{1 - L_{k_c(S)}(s)z^2}. \quad (7)$$

Indeed, for any remaining alphabet of size k ,

$$\sum_{m \geq 1} k^{m \bmod 2} L_k(s)^{\lfloor m/2 \rfloor} z^m = \frac{kz + L_k(s)z^2}{1 - L_k(s)z^2}.$$

Every denominator has constant term 1, so the identities are valid as formal power series. Summing over D is also valid coefficient by coefficient because $D \leq n$.

For example, $H_3(s) = 4 + s$: four matrices form singleton orbits, and the remaining two form the orbit with column sizes $(1, 2)$ and second-column record $\{1, 2\}$. In general,

$$\sum_{P \in \text{PM}_n} x^{p(P)} y^{q(P)} = H_n(x + y), \quad H_n(2) = n!,$$

where the last identity follows from the CDK bijection and the $1 \cdot 2 \cdots n$ choices of an inversion sequence. The original image, its two one-sided enlargements, and the full distribution of proper ascents and descents are therefore the zero-, one-, and two-orientation counts of the same family of orbits.

References

- [1] A. Claesson, M. Dukes, and M. Kubitzke, *Partition and composition matrices*, J. Combin. Theory Ser. A **118** (2011), 1624–1637. doi:10.1016/j.jcta.2011.02.001.
- [2] S. Chern and S. Fu, *Signed counting of partition matrices*, J. Combin. Theory Ser. A **223** (2026), Article 106213. arXiv:2508.21318v2, Definitions 2.1–2.2 and Question 5.5.
- [3] EULER case package, *The CDK image of improper partition matrices*, source and proof.